LECTURE NOTES

Lecture 7: Second-Order Methods

University

Simon Holm

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DEPARTMENT OF MATHEMATICS
AND COMPUTER SCIENCE

Contents

Derivative-Free Methods

- Also called direct search methods, zero-order, black box, pattern search
- Direct method search using function evaluations only

Cyclic Coordinate Search

Powell's Method

Hooke-Jeeves

evaluates $f(x)$ anf $f(x \pm \alpha e_i)$ for a given α in every coordinate direction from an anchoring point x

- It accepts any improvement it may find.
- If no improvements are found, it decreases the step size.
- The process repeats until the step size is sufficiently small.
- 2n evaluations for an n-dimensional problem

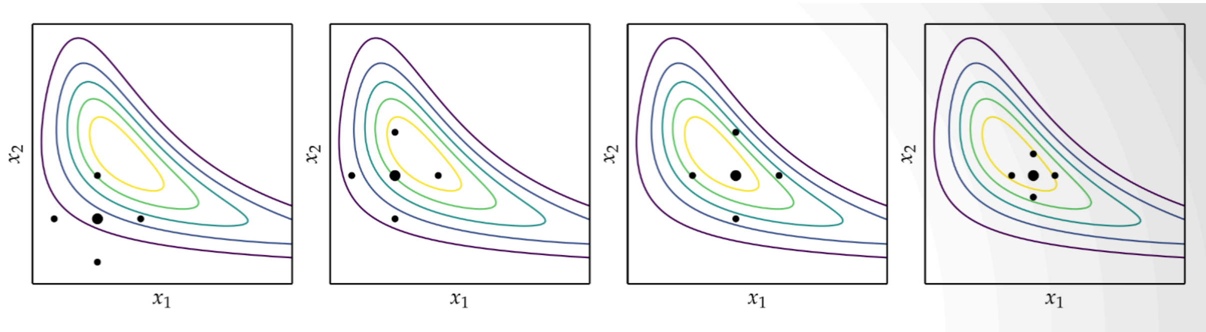


Figure 1:

Generalized Pattern Search

Generalization of Hooke-Jeeves method

It seaches for the best directions.

Construct a pattern P from set of directions $\mathcal{D}$ of size m

Note that you need $m + 1$ directions to span the space

$$P = \{x + \alpha d \text{ for each } d \in \mathcal{D}\}$$

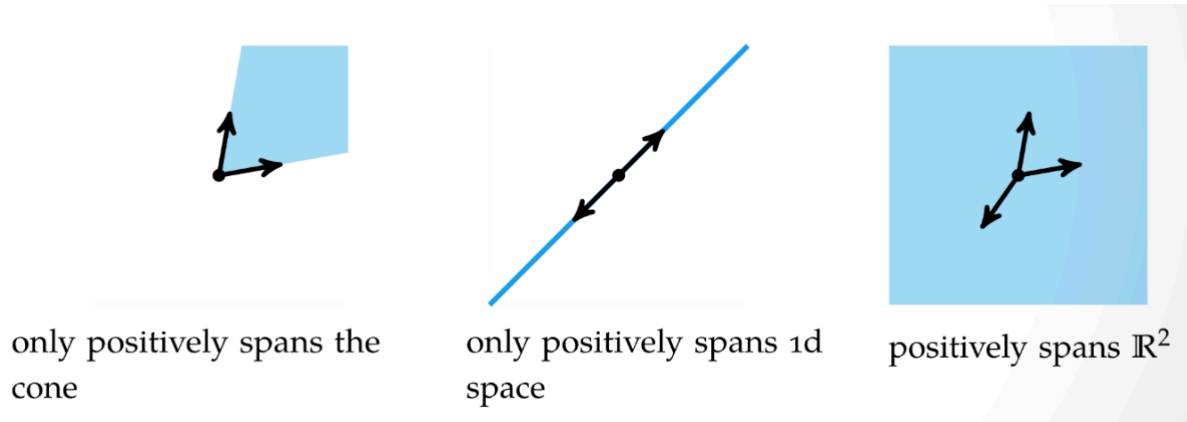


Figure 2: Vizualtion that we need 3 directions to span the space $\mathbb{R}^2$

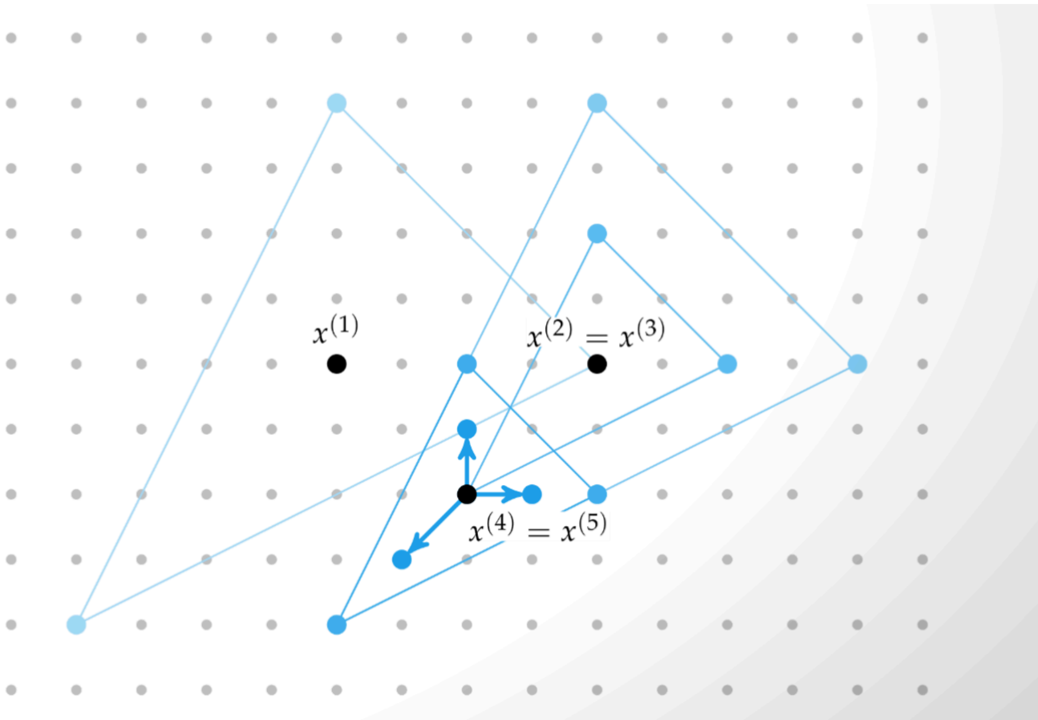


Figure 3: Example of Generalized Pattern Search

Nelder-Mead Simplex Method

Uses simple algorithm to traverse search space using set of points defining a simplex

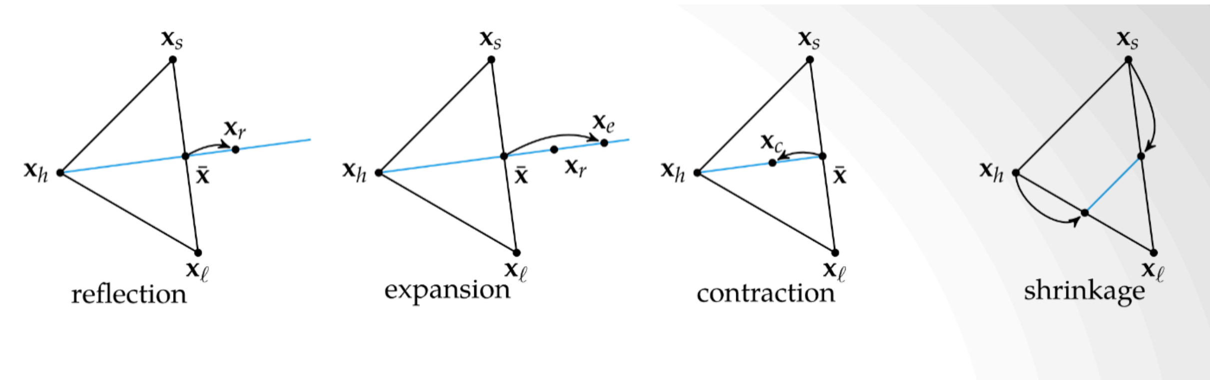
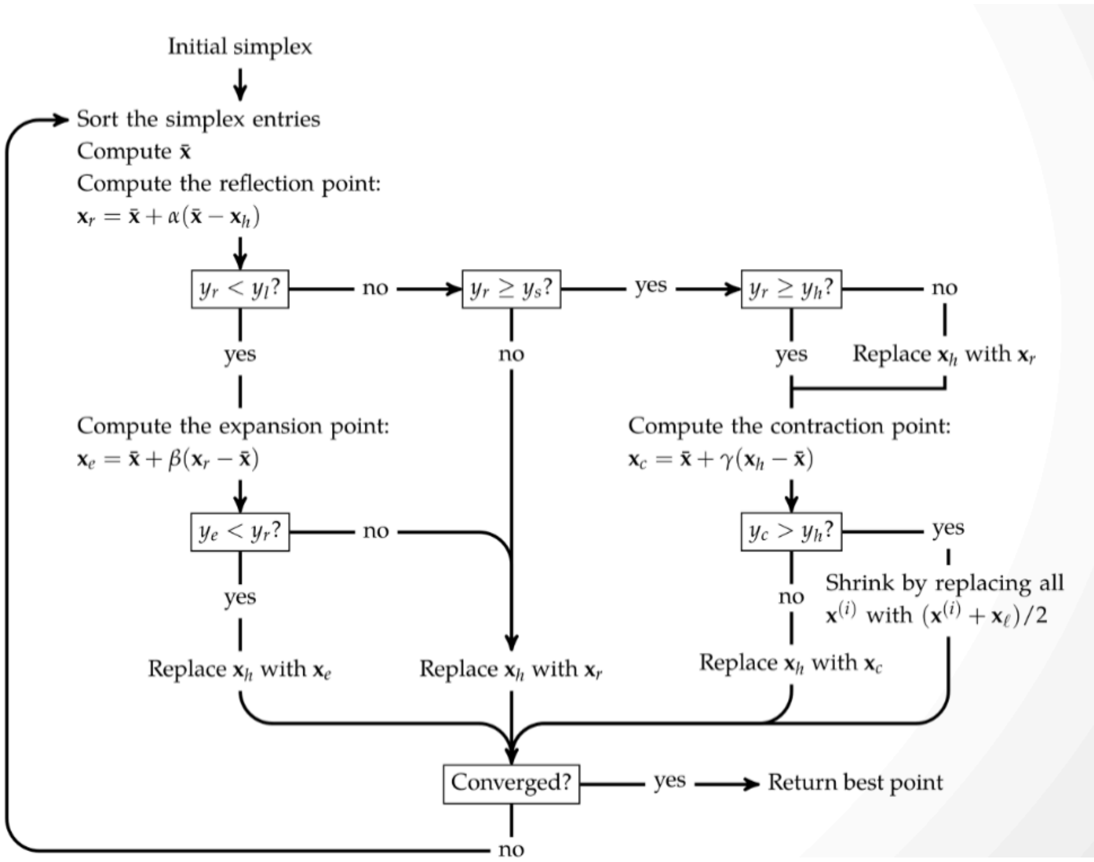


Figure 4: Nelder-Mead Simplex Method



, caption: [Nelder-Mead Simplex Method algorithm])

Algorithm Simplex search (Nelder-Mead)

Let $x_1 ... x_{n+1}$ be vertices of a simplex

Let $h : y_h = \max_i y_i = \max f(x_i)$ and $l : y_l = \min_i y_i = \min f(x_i)$

Let $|x|$ be the centroid of the points with $i \neq h$ and $d(x_i, x_j)$ the distance between two points x_i and x_j ($j \leftarrow 0$)

iter. $k : (k \leftarrow k + 1)$ Generate the **reflextion** x_R of x_h

Case 1 **if** $y_l \leq y_R < y_h$ **then** go to **Reflect**

Case 2 **else if** $y_R < y_l$ **then**

  Generate the **expantion** x_E of x_h

  if $y_E < y_l$ **then** $x_h \leftarrow x_E$ and go to **Reflect**

  else $x_h \leftarrow x_R$ and go to **Reflect**

Case 3 **else if** $y_R > y_i, \forall i \neq h$ **then** $x_h \leftarrow \min\{y_h, y_R\}$ and generate the **contraction 1**

  if $y_C \leq \min\{y_h, y_R\}$ **then** $x_h \leftarrow x_C$ and go to **Reflect**

  else contraction 2 $x_i \leftarrow \frac{x_i + x_l}{2}$

DIRECT - Dividet Rectangles

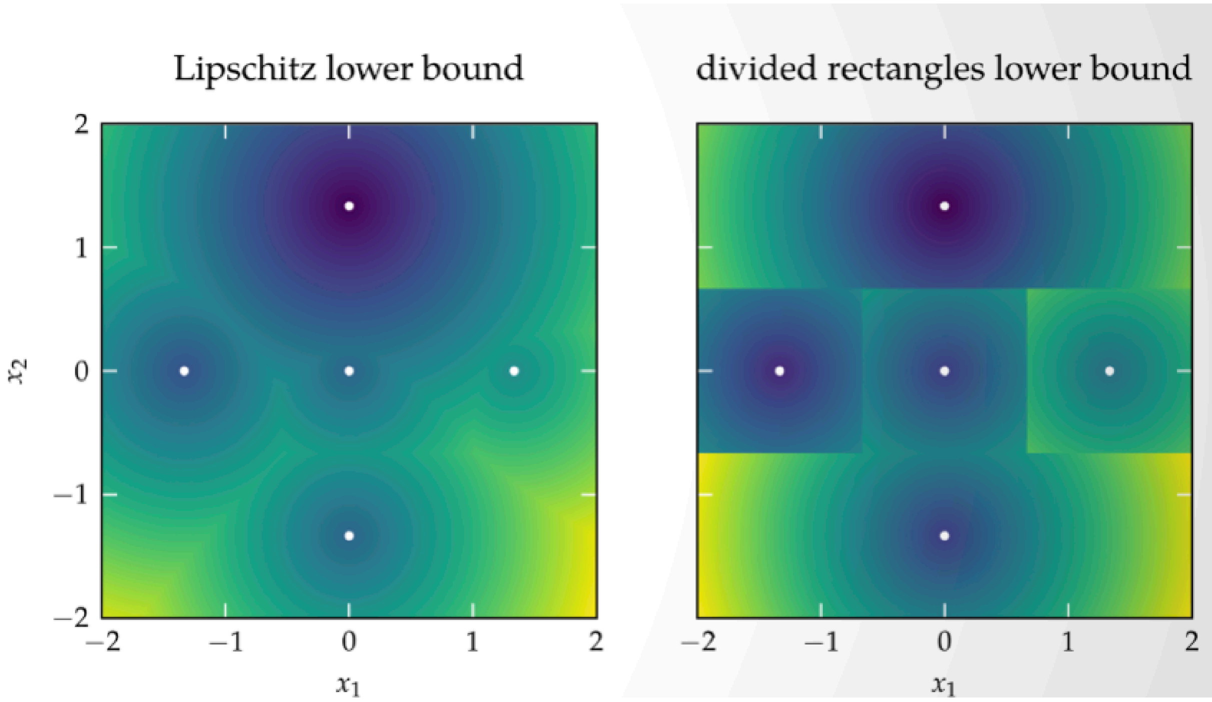


Figure 6:

Since we for blackbox problems we might now know anything about L

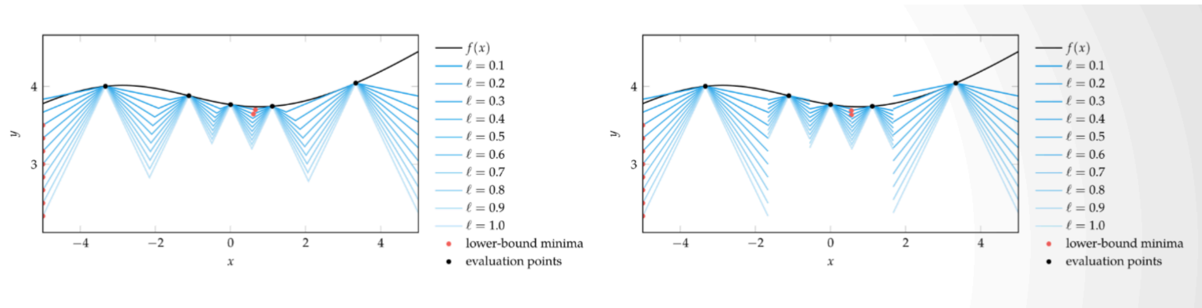


Figure 7:

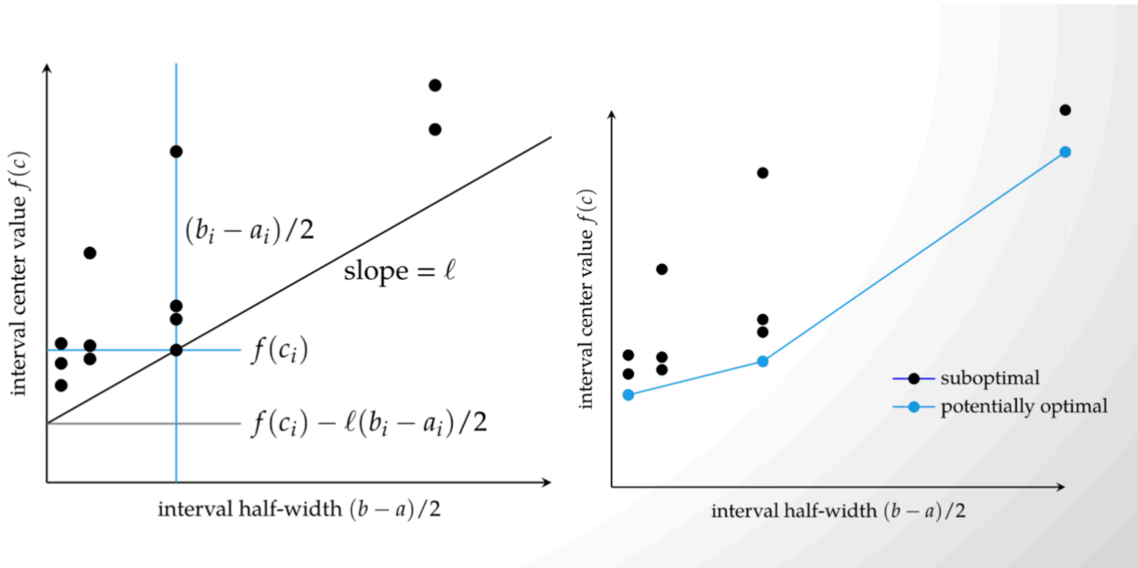


Figure 8:

Choose the largest

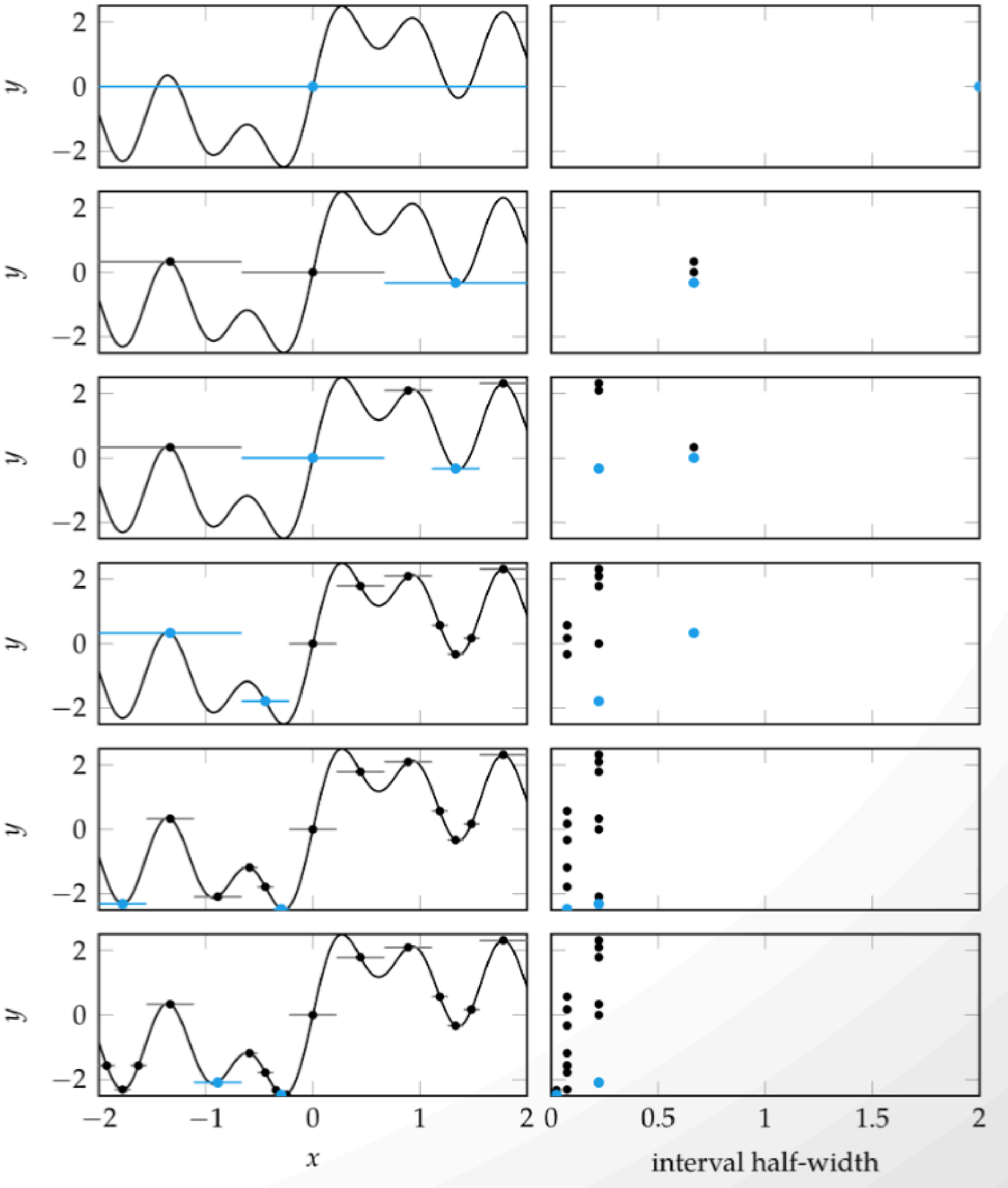


Figure 9:

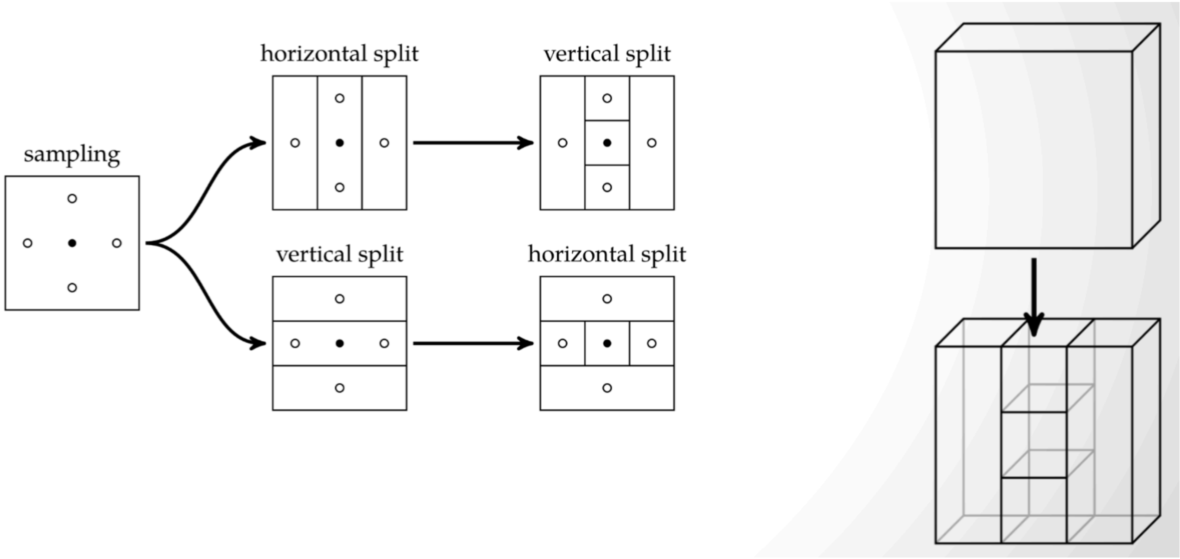


Figure 10:

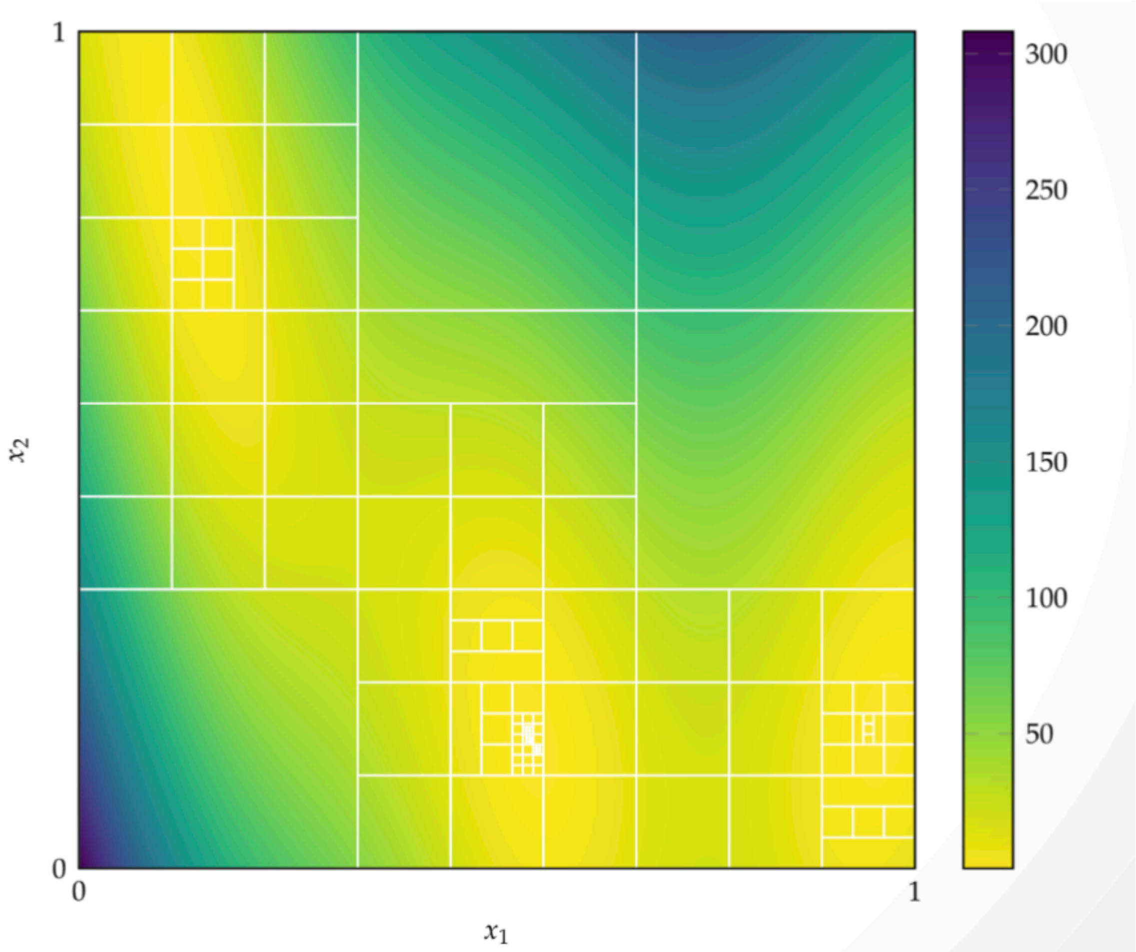


Figure 11: